

期中练习

一. 填空题(20分)

1. 298 K时, $\frac{1}{2} \text{SO}_2(\text{g}) = \frac{1}{2} \text{S}(\text{s}) + \frac{1}{2} \text{O}_2(\text{g}) \quad \Delta H_m^\ominus = 148.4 \text{ kJ.mol.}$ $\text{SO}_2(\text{g})$ 的标准生成焓为 $-296.8 \text{ kJ.mol}^{-1}$.

$$-\Delta H_m^\ominus = \Delta_f H_m^\ominus(\text{SO}_2, \text{g})$$

2. 在 100 kPa 下, 将 1 mol 单原子分子理想气体从 100 °C 加热到 200 °C, $Q =$ 2079 J.

$$1 \times \frac{5}{2} R \times 100 = 2078.5 \text{ J}$$

$$Q = \int_{T_1}^{T_2} n C_{p,m} dT = \frac{5}{2} R \times (200 - 100) = 2078.5 \text{ J}$$

3. 工作在 600 K 与 400 K 之间的热机, 其热机效率 η $\leq 33.3\%$, 若该机器每循环操作一次, 从高温热源吸热 10 kJ, 则最多可对外做功 3.33 kJ.

$$\eta \leq \eta_r = 1 - \frac{T_2}{T_1} = 1 - \frac{400}{600} = 0.333$$

$$\eta = -\frac{w}{Q_1}; \quad -w = \eta Q_1 = 0.333 \times 10 \text{ kJ} = 3.33 \text{ kJ}$$

4. 300 K 下, 1 mol H_2 (可看作理想气体) 由 100 kPa 压缩至 200 kPa 终态体积 $V =$ 12.5 dm³。

$$V_2 = \frac{nRT}{P_2} = \frac{1 \times 8.314 \times 300}{200} = 12.5 \text{ dm}^3$$

5. 1 mol N_2 气(可看作理想气体)由 303.9 kPa、25 °C 等温可逆膨胀到 24.45 dm^3 , 此过程的 $Q = \underline{\hspace{2cm} 2721 \text{ J} \hspace{2cm}}$, $\Delta U = \underline{\hspace{2cm} 0 \hspace{2cm}}$ 。

$$\Delta U = 0; \quad Q = -w = nRT \ln \frac{V_2}{V_1}$$

$$V_1 = \frac{nRT}{P_1} = 8.15 \times 10^{-3} \text{ m}^3$$

$$= 8.15 \text{ dm}^3$$

$$Q = -W = nRT \ln \frac{V_2}{V_1}$$

6. 甲苯在正常沸点 110 °C 时的气化焓为 362 J/g, 摩尔质量为 92 g/mol。1 mol 甲苯在正常沸点下蒸发为气体,

$$\Delta G = \underline{\hspace{2cm} 0 \hspace{2cm}}。$$

$$Q = \Delta H_m$$

$$\Delta S = \frac{Q}{T}$$

$$\Delta G = \Delta H - T\Delta S$$

7. 在 $-5\text{ }^{\circ}\text{C}$, 101.3 kPa 条件下, 1 mol 过冷水凝结成同样条件下的冰, 则系统及环境熵变应当为 $\Delta S(\text{系})$ < 0, $\Delta S(\text{环})$ > 0 (填 $>$, $<$ 或 $=$). 液 $\rightarrow$ 固 混乱度 $\downarrow$, $\Delta S < 0$.

$= -\frac{\Delta H_f}{T}$ 水 $\rightarrow$ 冰, 放热, $\Rightarrow \Delta S < 0$

8. 2 mol 单原子分子理想气体, 从 $0\text{ }^{\circ}\text{C}$, 1 MPa 绝热不可逆膨胀到 0.1 MPa , 并对外做功 0.1 kJ , 此过程的 $\Delta U =$ -0.1 kJ 0. $\Delta = 0$

$\Delta U =$ -0.1 kJ 0.

$\Delta U = W =$

$$Q = 0; \quad \Delta U = w = -0.1\text{ kJ}$$

9. 100 °C 和 101.3 kPa 下的 1 mol $\text{H}_2\text{O}(\text{l})$ 与 100 °C 的大热源相接触, 并使其向真空器皿中蒸发为 100 °C、101.3 kPa 的水蒸气 $\text{H}_2\text{O}(\text{g})$. 对于这一过程, $\Delta S(\text{系})$ 、 ΔU 、 ΔG 和 $\Delta S(\text{总})$ 四个热力学量中可判断过程方向的热力学量是 $\Delta S(\text{总})$.

10. 1 mol 理想气体在 25 °C 时恒温可逆地从 1 dm^3 膨胀到 10 dm^3 , 系统的 $\Delta S =$ 19.14 J/K。

$$\Delta S = nR \ln \frac{V_2}{V_1} = 1 \times 8.314 \times \ln 10 = 19.14 \text{ J} \cdot \text{K}^{-1}$$

$$\Delta S = nR \ln \frac{V_2}{V_1} = 1 \times 8.314 \times \ln 10$$

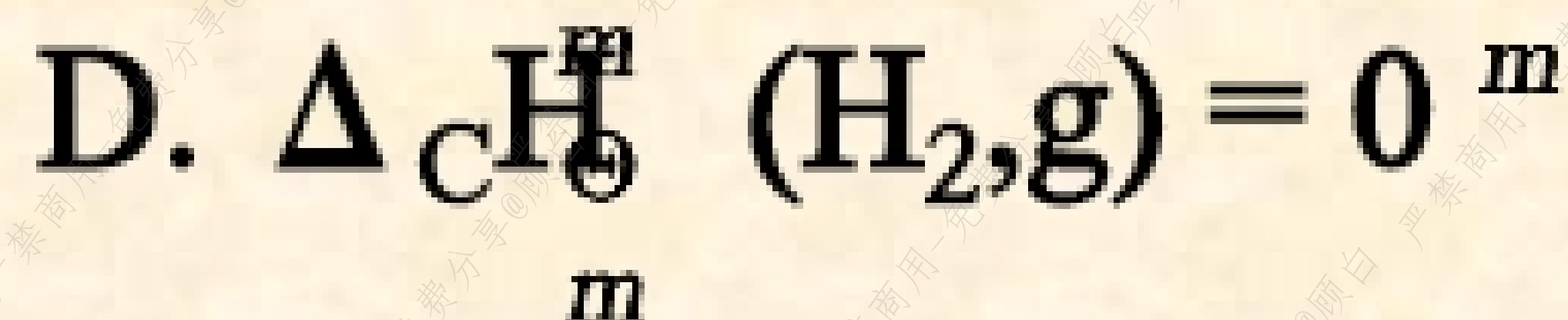
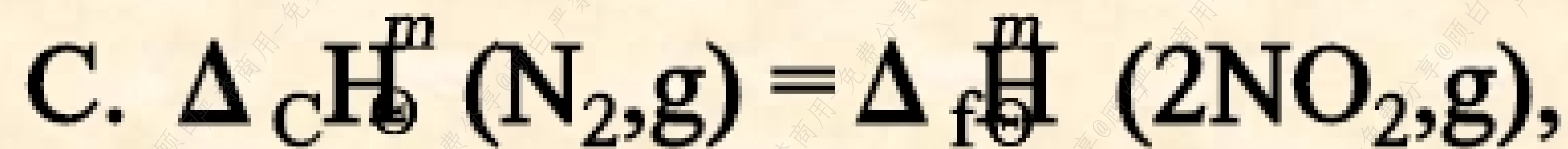
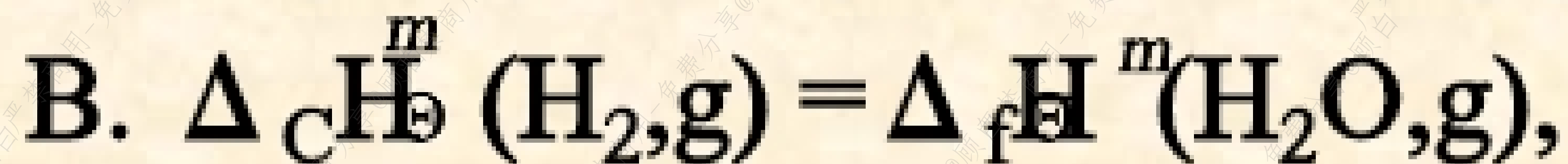
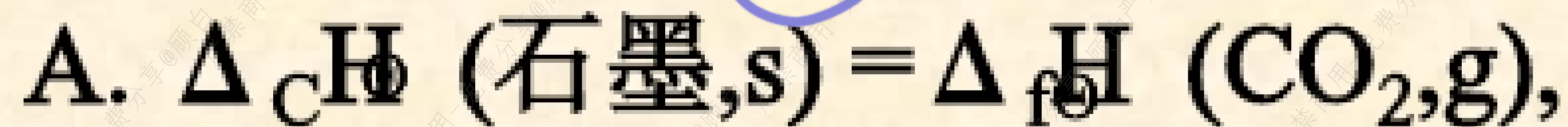
二.选择题(20分)

1. (2分) 1 mol 水在 100℃, 101.3 kPa 下变成同温同压下的水蒸气(水蒸气视为理想气体), 然后等温可逆膨胀到 10.13 kPa, 则全过程的 $\Delta H =$ C (已知水的气化焓为 40.67 kJ/mol).

A. 0, B. 40.67 J, C. 40.67 kJ, D. - 40.67 J

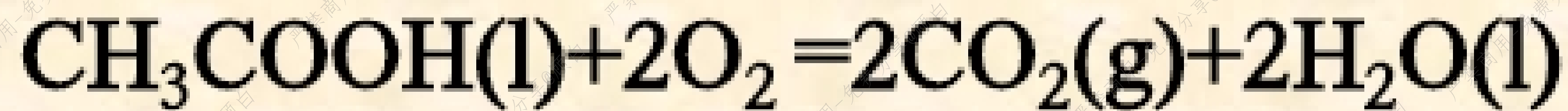
2.(2分)

下列关系式中正确的是: A



3. 已知 $\text{CH}_3\text{COOH}(\text{l})$, $\text{CO}_2(\text{g})$ 及 $\text{H}_2\text{O}(\text{l})$ 的标准生成焓 $\Delta_f H^\ominus_m$ (298K)/(kJmol⁻¹)分别为 -484.5, -393.5及-285.8, 则 $\text{CH}_3\text{COOH}(\text{l})$ 的标准燃烧焓

$\Delta_c H^\ominus_m$ (l, 298K) = C kJmol⁻¹ $\text{CH}_3\text{COOH}(\text{l}) + 2\text{O}_2 = 2\text{H}_2\text{O}(\text{l}) + 2\text{CO}_2$
A. 484.5, B. 0 C. -874.1 D. 874.1 $\Delta_c H^\ominus_m = -393.5 \times 2$
 $- 285.8 \times 2$
 $- (-484.5)$
 $= -874.1 \text{ kJ/mol}$



4. H₂的标准燃烧焓等于

B

A. H₂的标准生成焓,

B. H₂O(l)的标准生成焓,

C. O₂的标准生成焓,

D. 零.

5. 在 0°C , 101.3 kPa 下, 1 mol 水凝结成同样条件下的冰, 则系统, 环境及总熵变应该是 C .

- A. $\Delta S(\text{系}) < 0$, $\Delta S(\text{环}) < 0$, $\Delta S(\text{总}) < 0$;
- B. $\Delta S(\text{系}) > 0$, $\Delta S(\text{环}) > 0$, $\Delta S(\text{总}) > 0$;
- C. $\Delta S(\text{系}) < 0$, $\Delta S(\text{环}) > 0$, $\Delta S(\text{总}) = 0$;
- D. $\Delta S(\text{系}) < 0$, $\Delta S(\text{环}) > 0$, $\Delta S(\text{总}) > 0$.

6.(2分)

在标准压力 p^{θ} 下, 383 K 的水变为同温度下的蒸气, 吸热 Q_p 此相变过程中, 不能成立的关系式为 C .

- A. $\Delta G < 0$, B. $\Delta H = Q_p$ C. $\Delta S \leq 0$, D. $\Delta S > 0$

$= \Delta H - T\Delta S$

自发, $\Delta G < 0$

7.(2分)

对一封闭的绝热系统做功后,系统的温度 B .
A. 不变, B. 升高, C. 降低, D. 难以确定.

8.(2分)

100℃, 101.3 kPa下的1 mol水与 100℃热源接触,使它向真空中蒸发成 100℃, 101.3 kPa的水蒸气,此过程不正确的是 C

C
A. $\Delta H > 0$, B. $\Delta U > 0$, C. $\Delta G \leq 0$, D. $\Delta S > 0$

吸热

$$Q = \Delta H > 0$$

$\Delta V \uparrow$

$$\Delta U = \underbrace{\Delta H}_{+} - \underbrace{p\Delta V}_{+}$$

9.(2分)

101.3 kPa下,110 °C的水变为110 °C的蒸气,吸热 Q .此过程环境的熵变 $\Delta S(\text{环})$ C .

A. = 0, B. > 0, C. < 0, D. = Q/T .

10.(2分)

某系统内初态 A 经几种不同的可逆途径到达相同的终态 B, 经各途径的系统熵变 ΔS A .

A.均相同, B.各不相同,
C.不一定相同, D.都不等于经不可逆途径到达相同终态的熵变.

选择题答案：

- 1、C
- 2、A
- 3、C
- 4、B
- 5、C
- 6、C
- 7、B
- 8、C
- 9、C
- 10、A

三. (15分)

10dm³He (可看作理想气体) 从20℃,

1MPa经下列不同过程膨胀到终态压力为

100kPa: (1) 等温可逆膨胀; (2) 绝热

可逆膨胀。分别计算各过程的W, Q, ΔU和

ΔH。

$$n = \frac{PV}{RT} = \frac{1 \times 10^6 \times 10 \times 10^{-3}}{8.314 \times 293.15} = 4.11 \text{ mol}$$

$$Q_2 = 0$$

41).

$$Q = -nRT \ln \frac{P_2}{P_1} = 23.1 \text{ kJ}$$

$$\gamma = \frac{C_{p,m}}{C_{v,m}} = \frac{5}{3}$$

$$\Delta U = W = n C_{v,m} \Delta T$$

$$\frac{T_2}{T_1} = \left(\frac{P_1}{P_2}\right)^{\frac{1}{\gamma}}$$

$$\frac{T_2}{293} = \left(\frac{1}{10}\right)^{\frac{3}{5}} \Rightarrow T_2 = 116.5 \text{ K}$$

$$W = nRT \ln \frac{P_2}{P_1} = 4.11 \times 8.314 \times 293 \times \ln \frac{10^5}{10^6} = -23.1 \text{ kJ}$$

$$\Delta U = W + Q = 0$$

$$W_2 = \Delta U_2 = 4.11 \times \frac{5}{2} \times 8.314 \times (116.5 - 293) = -9047 \text{ J}$$

$$\Delta H = 0$$

$$\Delta H_2 = n C_{p,m} \Delta T = 4.11 \times \frac{5}{2} \times 8.314 \times (116.5 - 293) = -15078 \text{ J}$$

$$n = \frac{pV}{RT} = \frac{1 \times 10^6 \times 10 \times 10^{-3}}{8.314 \times 293} = 4.11 \text{ (mol)}$$

(1) $\Delta U_1 = \Delta H_1 = 0$ (理想气体恒温过程)

$$\begin{aligned} W_1 &= nRT \ln(p_2/p_1) \\ &= 4.11 \times 8.314 \times 293 \times \ln(10^5/10^6) \\ &= -23.1 \text{ kJ} \end{aligned}$$

$$Q_1 = -W_1 = 23.1 \text{ kJ}$$

(2) $Q_2=0$ (绝热)

$$\gamma = \frac{C_{p,m}}{C_{v,m}} = \frac{5}{3} = 1.67 \quad T_2 = \left(\frac{p_1}{p_2} \right)^{\frac{1-\gamma}{\gamma}} T_1 = 116.5 K$$

$$\begin{aligned} W_2 &= \Delta U_2 = nC_{v,m}(T_2 - T_1) \\ &= 4.11 \times 1.5 \times 8.314 \times (116.5 - 293) \\ &= -9047 \text{ J} \end{aligned}$$

$$\begin{aligned} \Delta H_2 &= nC_{p,m}(T_2 - T_1) \\ &= 4.11 \times 2.5 \times 8.314 \times (116.5 - 293) \\ &= -15078 \text{ J} \end{aligned}$$

四. (15分)

$$\Delta_r H = \Delta_c H(C_2H_2O_4, s) = -251 \text{ kJ/mol.}$$

$$= 2\Delta_f H(CO_2, g) + \Delta_f H(H_2O, l) - \Delta_f H(H_2C_2O_4, s).$$

$$\Rightarrow \Delta_f H(H_2C_2O_4) = 2 \times (-393.5) - (-285.9) - (-251) = -821.1 \text{ kJ/mol}$$



298K时草酸($H_2C_2O_4$)的标准燃烧焓为 -251 kJ/mol :

(2) $2C(s) + H_2(g) + 2O_2(g) \rightarrow H_2C_2O_4(s)$ $\Delta_r H = \Delta_f H(H_2C_2O_4)$ $\sum \nu_B C_p(B)$

(1) 计算298K时草酸的标准生成焓;

(2) 计算398K时草酸的标准生成焓;

(3) 计算298K时草酸(固体)燃烧时的 ΔU .

$$= C_p(\text{草}) - 2C_p(\text{石墨}) - C_p(H_2, g) - 2C_p(O_2, g)$$

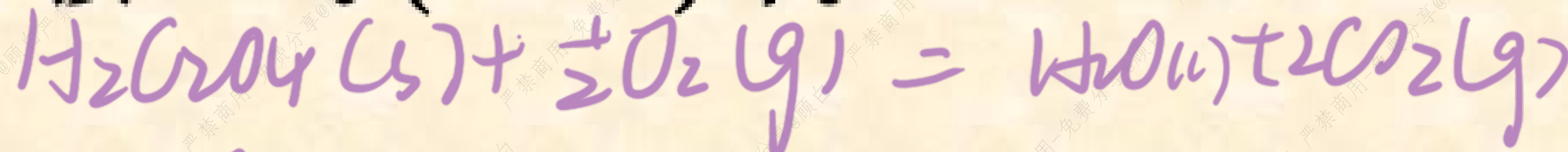
$$= 97.91 - 2 \times 8.79 - 29.0 \times 3 = -822.6 \text{ kJ/mol}$$

已知298K时 $H_2O(l)$ 及 $CO_2(g)$ 的标准生成焓分别为

-285.9 kJ/mol 和 -393.5 kJ/mol , $C(\text{石墨})$ 和草酸(固)的 C_{pm}

分别为 8.79 J/(K.mol) 和 97.91 J/(K.mol) 。气体的 C_{pm} 均

按 29.0 J/(K.mol) 计。



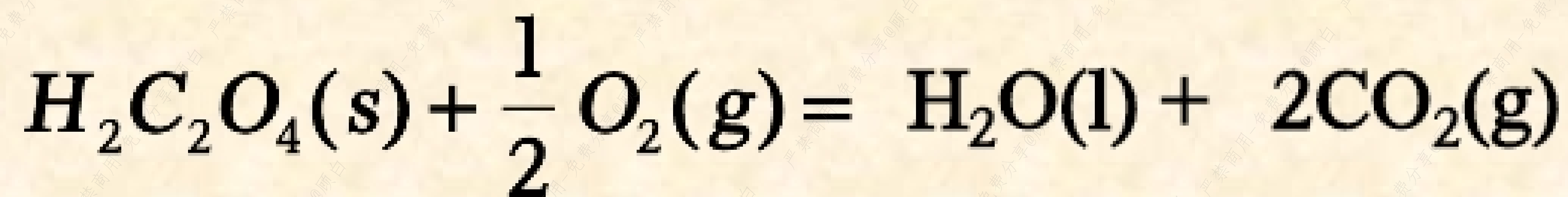
$$\Delta_r H_m^\ominus = -251 \text{ kJ/mol}$$

$$\Delta U = \Delta_r H_m - \sum \nu_B(g) RT = -251 - 1.5 \times 8.314 \times 298 \times 10^{-3}$$

$$\Delta H = \Delta U + \Delta n RT \quad \Delta n = 2 - \frac{1}{2} = \frac{3}{2}$$

$$= -254.7 \text{ kJ/mol}$$

解：有关化学反应的题目先写出反应方程式



298K:

$$\Delta_{\text{r}}H = \Delta_{\text{c}}H(\text{H}_2\text{C}_2\text{O}_4, \text{s}) = -251 \text{ kJ/mol}$$

$$\Delta_{\text{r}}H = \Delta_{\text{c}}H(\text{H}_2\text{C}_2\text{O}_4, \text{s})$$

$$= 2 \Delta_{\text{f}}H(\text{CO}_2, \text{g}) + \Delta_{\text{f}}H(\text{H}_2\text{O}, \text{l}) - \Delta_{\text{f}}H(\text{H}_2\text{C}_2\text{O}_4, \text{s})$$

$$\Delta_{\text{f}}H(\text{H}_2\text{C}_2\text{O}_4, \text{s})$$

$$= 2 \Delta_{\text{f}}H(\text{CO}_2, \text{g}) + \Delta_{\text{f}}H(\text{H}_2\text{O}, \text{l}) - \Delta_{\text{c}}H(\text{H}_2\text{C}_2\text{O}_4, \text{s})$$

$$= 2 \times (-393.5) - 285.9 + 251 = -821.9 \text{ kJ/mol}$$

(2) 398K



$$\Delta_r H = \Delta_f H(\text{H}_2\text{C}_2\text{O}_4)$$

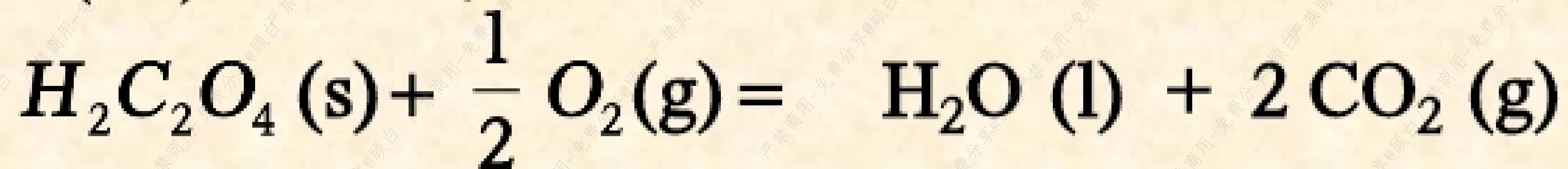
$$\sum \nu_B C_p(B) = C_p(\text{草酸}) - 2C_p(\text{石墨}) - C_p(\text{H}_2, \text{g}) - 2C_p(\text{O}_2, \text{g})$$

$$= 97.91 - 2 \times 8.79 - 3 \times 29.0 = -6.67 \text{ J.K}^{-1}.\text{mol}^{-1}$$

$$\Delta_r H(398\text{K}) = \Delta_r H(298\text{K}) + \sum \nu_B C_p(B) (T_2 - T_1)$$

$$= -821.9 + (-6.67) \times (398 - 298) \times 10^{-3} = -822.6 \text{ kJ/mol}$$

(3) 298K:



$$\Delta_r H_m^\ominus = -251 \text{ kJ} / \text{mol}$$

$$\Delta_r U_m = \Delta_r H_m - \sum \nu_{B(\text{g})} RT$$

$$\Delta U = -251 - 1.5 \times 8.314 \times 298 \times 10^{-3} = -254.7 (\text{kJ} / \text{mol})$$

五. (15分)

1mol单原子分子理想气体恒定在101.3kPa下从200 °C加热到400 °C。已知200 °C时该气体的标准熵为135.1J.K⁻¹.mol⁻¹。计算此过程的 Q, W, ΔU, ΔH, ΔS 及 ΔG。

~~W ≠ 0~~ 反抗恒外压. 非恒压. $W = p \Delta V = -\Delta nRT$

$$\Delta U = n C_{V,m} \Delta T = 1 \times \frac{3}{2} \times 8.314 \times 200 = -1663 \text{ J}$$

$$\Delta H = n C_{p,m} \Delta T = 1 \times \frac{5}{2} \times 8.314 \times 200 = 4157 \text{ J}$$

$$\Delta S = n C_{p,m} \ln \frac{T_2}{T_1} = 1 \times \frac{5}{2} \times 8.314 \ln \frac{673.15}{473.15} = 7.33 \text{ J/K}$$

$$S(673K) = S(473K) + \Delta S = 135.1 + 7.33 = 142.4 \text{ J/K}$$

$$\Delta G = \Delta H - \Delta TS = \Delta H - (T_2 S_2 - T_1 S_1) = 4157 - (673 \times 142.4 - 473 \times 135.1) \approx -27.8 \text{ kJ}$$

$$\Delta A = \Delta U - \Delta TS = 2494 - (673 \times 142.4 - 473 \times 135.1) \approx -29.4 \text{ kJ}$$

五. 解 分清反抗恒外压和恒压过程的区别

$$W = -p\Delta V = -\Delta(pV) = -\Delta nRT$$

$$= -1 \times 8.314 \times (400 - 200) = -1663 \text{ J}$$

$$\Delta U = \int_{T_1}^{T_2} nC_{V,m} dT = 1 \times \frac{3}{2} \times 8.314 \times (400 - 200) = 2494 \text{ J}$$

$$\Delta H = \int_{T_1}^{T_2} nC_{p,m} dT = 1 \times \frac{5}{2} \times 8.314 \times (400 - 200) = 4157 \text{ J}$$

$$\Delta S = nC_{p,m} \ln \frac{T_2}{T_1} = \frac{5}{2} \times 8.314 \times \ln \frac{673.15}{473.15} = 7.33 \text{ J / K}$$

$$Q = \Delta H = 4157 \text{ J}$$

$$S(673\text{K}) = S(473\text{K}) + \Delta S = 135.1 + 7.33 = 142.4 \text{ J.K}^{-1}$$

$$\begin{aligned}\Delta G &= \Delta H - (T_2 S_2 - T_1 S_1) \\ &= 4157 - (673 \times 142.4 - 473 \times 135.1) \\ &= -27776 \text{ J} \approx -27.8 \text{ kJ}\end{aligned}$$

$$\begin{aligned}\Delta A &= \Delta U - (T_2 S_2 - T_1 S_1) \\ &= 2492 - (673 \times 142.4 - 473 \times 135.1) \\ &= -29441 \text{ J} \approx -29.4 \text{ kJ}\end{aligned}$$

六. (15分)

在 25 °C 将 1mol 101.3kPa 的 O_2 (理想气体) 在 恒定外压 607.8kPa 的条件下压缩到达平衡, 求此过程的 Q , W , ΔU , ΔH , ΔG , $\Delta S(\text{系})$ 及 $\Delta S(\text{总})$ 。

$$\Delta U = \Delta H = 0.$$

$$W = -Q = P_{\text{外}}(V_2 - V_1)$$

$$= P_{\text{外}} \left(\frac{RT}{P_2} - \frac{RT}{P_1} \right) = 607.8 \times 8.314 \times 298 \left(\frac{1}{607.8} - \frac{1}{101.3} \right) \approx -12.9 \text{ kJ}$$

$$\Delta S(\text{系}) = nR \ln \frac{P_1}{P_2} = 1 \times 8.314 \ln \frac{101.3}{607.8} = -14.9 \text{ J/K}$$

$$\Delta S(\text{环}) = -\frac{Q}{T} = \frac{12.9}{298}$$

$$\Delta S(\text{总}) = \Delta S(\text{系}) + \Delta S(\text{环})$$

六. 解

此过程为恒温压缩过程, $\Delta U = 0$ $\Delta H = 0$

$$\begin{aligned} -W &= Q = p(\text{外})(V_2 - V_1) = p(\text{外})(RT/p_2 - RT/p_1) \\ &= 607.8 \times 8.314 \times 298(1/607.8 - 1/101.3) \\ &= -12388 \text{ J} \approx -12.39 \text{ kJ} \end{aligned}$$

$$\begin{aligned} \Delta S(\text{系}) &= nR \ln(p_1/p_2) \\ &= 1 \times 8.314 \ln(101.3/607.8) = -14.90 \text{ J/K} \end{aligned}$$

$$\Delta S(\text{环}) = -Q/T = 12388/298 = 41.57 \text{ J/K}$$

$$\begin{aligned} \Delta S(\text{总}) &= \Delta S(\text{系}) + \Delta S(\text{环}) \\ &= -14.90 + 41.57 = 26.67 \text{ J/K} \end{aligned}$$

$$\Delta G = \Delta H - T \Delta S = 0 - 298 \times (-14.90) = 4440 \text{ J}$$